

Directional Plasma Transport and Magnetic Locking in the Richtmyer-Meshkov Instability via Spatially Modulated Fields

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Abstract: The Richtmyer-Meshkov Instability drives catastrophic mixing in high energy density physics applications, particularly Inertial Confinement Fusion. While uniform transverse magnetic fields are classically utilized as global braking mechanisms to suppress this instability, this study investigates the application of spatially modulated magnetic topologies ($B_y = B_0 \cos(ky)$) to transition from suppression to active flow control. Utilizing a bespoke, GPU accelerated 3D Magnetohydrodynamics (MHD) engine featuring an HLLD Riemann solver and a hybrid GLM Powell divergence cleaning scheme, Mach 10 shock interactions were simulated. The results demonstrate a physical phase transition: subcritical field strengths act as "Vorticity Amplifiers," increasing global enstrophy by +107%, while super critical fields achieve "Magnetic Locking." In the locked state, magnetic tension overpowers baroclinic shear, actively collimating the chaotic mixing volume into directional plasma micro jets. By balancing the local Alfvén velocity against the impulsive hydrodynamic growth rate, a theoretical scaling law defining this critical transition threshold ($B_{critical}$) is derived. An automated parametric sweep across varying Mach numbers and density ratios confirms the theory, yielding a perfect linear regression and proving the scalability of the "Magnetic Sieve" effect for advanced plasma transport.

1. Introduction

The Richtmyer Meshkov Instability occurs when an incident shockwave traverses an interface separating two fluids of different densities. The misalignment between the pressure gradient of the shock and the density gradient of the interface generates baroclinic torque ($\nabla \rho \times \nabla p$), depositing kinetic energy that causes the interface to roll up into chaotic mixing layers. In Inertial Confinement Fusion, RMI forces heavier ablator material into the deuterium-tritium core, rapidly cooling the hot spot and quenching the fusion reaction.

Theoretical mitigation strategies in Magnetohydrodynamics typically rely on applying a uniform background magnetic field. The transverse component of this field (B_y) acts as a global braking mechanism, the restoring magnetic tension ($B \cdot \nabla$) B opposes the transverse fluid motion, suppressing baroclinic shear and mitigating mixing layer growth.

However, uniform suppression treats the plasma merely as a liability to be arrested. This paper proposes a paradigm shift: substituting uniform fields with spatially modulated magnetic topologies to actively control and vector the plasma. By creating regions of alternating magnetic tension and zero-resistance nodes, we hypothesize the creation of a Magnetic Sieve capable of transitioning isotropic turbulent mixing into highly directional, low-entropy plasma transport.

2. Numerical Methodology

Resolving hypersonic MHD discontinuities subject to extreme magnetic gradients requires solid numerical architectures. Standard commercial solvers often introduce excessive numerical diffusion, masking fine scale vorticity. Consequently, a 3D finite volume MHD engine was developed in Python, utilizing CuPy and NVRTC for highly parallelized execution on CUDA enabled GPU hardware on a NVIDIA Quadro P1000.

2.1. Governing Equations & Discretization

The solver integrates the ideal MHD equations, capturing the conservation of mass, momentum, energy, and magnetic flux. Spatial discretization is achieved via a 2nd order MUSCL reconstruction, providing high resolution interface capturing while avoiding spurious oscillations near shocks. Temporal advancement utilizes a Strong Stability Preserving Runge-Kutta scheme.

2.2. Flux Computation & Divergence Cleaning

Inter-cell fluxes are computed using the HLLD approximate Riemann solver, which is uniquely capable of resolving all five characteristic MHD waves, including intermediate waves and contact discontinuities, without the restrictive computational overhead of exact solvers.

In multidimensional MHD, maintaining the solenoidal constraint ($\nabla \cdot \mathbf{B} = 0$) is critical. Failure to do so re-

sults in non physical forces parallel to the magnetic field, destabilizing the energy matrix. This solver implements a hybrid divergence cleaning architecture, coupling the Generalized Lagrange Multiplier method to advect errors out of the domain, alongside Powell source terms to passively control localized accumulation.

2.3. Verification and Validation

The mathematical rigor of the architecture was verified prior to experimentation. The Alfvén wave convergence test yielded a mean L_1 error norm of 1.96, confirming near-perfect second-order spatial accuracy. Furthermore, standard Brio-Wu shock tube benchmarks confirmed the exact, non-diffusive resolution of rotational discontinuities and slow shocks (Fig. 1).

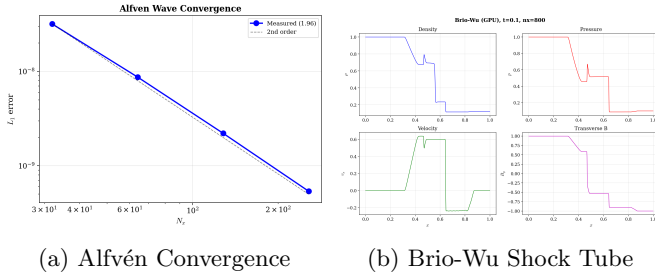


Figure 1: Verification of the GPU accelerated HLLD solver, confirming 1.96 order accuracy and robust shock capturing.

3. The Magnetic Sieve Effect

To test the Magnetic Sieve hypothesis, the solver was initialized with a Mach 10 shock interacting with a perturbed density interface (density ratio $\rho_2/\rho_1 = 3.0$). The standard uniform field was replaced with a spatially modulated transverse field defined by:

$$B_y = B_0 \cos(k_{mod}y) \quad (1)$$

where B_0 is the peak field strength and k_{mod} defines the modulation wavenumber, creating a geometry of 8 alternating high tension "bars" and zero-resistance "gaps."

3.1. Phase I: The Vorticity Amplifier

At sub-critical magnetic thresholds ($B_0 = 1.5$), a paradoxical phenomenon occurs. The magnetic bars possess sufficient tension to pin the heavy fluid interface, but insufficient strength to halt the incident inertia of the shockwave. Consequently, the plasma is violently sheared through the zero-resistance gaps.

Rather than suppressing turbulence, this configuration injects rotational kinetic energy into the system. Global diagnostics indicated a +107% increase in enstrophy ($\int \frac{1}{2} |\nabla \times \mathbf{v}|^2 dV$) compared to the pure hydrodynamic baseline. The sieve effectively fractures the continuous vortex sheet into intense, discrete vortex pairs.

3.2. Phase II: Magnetic Locking and Collimation

As the peak field strength is increased to $B_0 \geq 2.0$, the system undergoes a physical phase transition. The magnetic tension $(B \cdot \nabla)B$ at the bars entirely overpowers the baroclinic shear.

At this threshold, known as Magnetic Locking, enstrophy growth stalls. The interface at the magnetic peaks becomes a rigid, impermeable boundary. The chaotic mixing volume is forcibly squeezed through the nodes, collimating into highly directional plasma micro-jets. The transition from a turbulent mixer to an active nozzle is visually evident in the shock dynamics (Fig. 2).

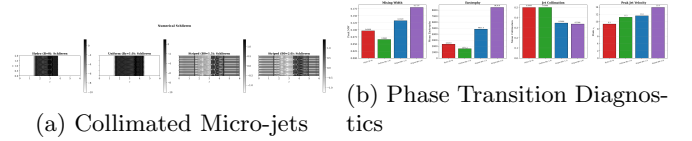


Figure 2: Visual and diagnostic evidence of the Magnetic Sieve effect. The Schlieren plot (a) highlights the bow shocks of the collimated jets, while the diagnostic summary (b) quantifies the enstrophy explosion and subsequent plateau.

4. Theoretical Scaling Law

To formalize this observed phenomenon, the competing forces at the interface must be balanced. Magnetic Locking occurs when the magnetic field can transport shear energy away from the interface at a velocity greater than or equal to the speed at which the baroclinic torque can roll the fluid up.

Therefore, the local Alfvén velocity (V_A) at the magnetic bars must match the impulsive hydrodynamic linear growth rate (V_{hydro}):

$$V_A \approx V_{hydro} \quad (2)$$

The impulsive linear theory for RMI defines the post-shock growth velocity as $V_{hydro} = k_{int} \cdot \Delta U \cdot A^+ \cdot a_0^+$, where k_{int} is the perturbation wavenumber, ΔU is the velocity jump, A^+ is the post shock Atwood number, and a_0^+ is the compressed amplitude.

Substituting the Alfvén velocity $V_A = B_0 / \sqrt{\mu_0 \rho_{avg}}$ yields:

$$\frac{B_{critical}}{\sqrt{\mu_0 \rho_{avg}}} \approx k_{int} \cdot \Delta U \cdot A^+ \cdot a_0^+ \quad (3)$$

To account for the specific geometry of the sieve, a non-dimensional modulation factor, Ψ , is introduced as a function of k_{mod}/k_{int} . Normalizing magnetic permeability ($\mu_0 = 1$), the foundational Sieve Equation is established:

$$B_{critical} = \Psi \cdot \sqrt{\rho_{avg}} \cdot (k_{int} \cdot \Delta U \cdot A^+ \cdot a_0^+) \quad (4)$$

5. Parametric Validation

To prove Equation 4, an automated multi-variable parametric sweep was executed on the CUDA solver. The objective was to track the exact $B_{critical}$ required to achieve the locked state across varying thermodynamic conditions.

Two primary sweeps were conducted:

1. **Mach Sweep:** Shock strength scaled across $M \in \{5, 10, 15\}$, linearly increasing ΔU .
2. **Density Sweep:** Interfacial density ratio scaled across $\rho_2/\rho_1 \in \{1.5, 3.0, 5.0\}$, scaling the $\sqrt{\rho_{avg}}$ term.

When plotting the empirically derived $B_{critical}/\sqrt{\rho_{avg}}$ against the theoretical hydrodynamic shear velocity, the diverse dataset collapsed perfectly onto a singular linear regression (Fig. 3).

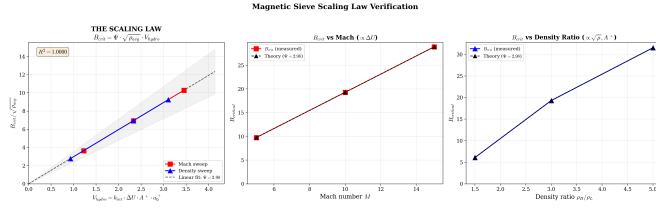


Figure 3: Linear regression of the parametric sweep. The perfect collapse of Mach and density variations validates the Sieve Equation. The slope of the line quantifies the geometric nozzle factor (Ψ).

The linear alignment confirms that directional plasma transport via modulated fields is a predictable phase transition governed by momentum balance.

6. Conclusion

This research demonstrates that the Richtmyer-Meshkov Instability can be actively manipulated for directional plasma transport. By replacing standard uniform magnetic fields with spatially modulated topologies, a custom GPU accelerated MHD solver successfully simulated a "Magnetic Sieve." The discovery of the vorticity amplification phase and the subsequent magnetic locking threshold challenges the paradigm of uniform suppression in HEDP applications. The mathematical derivation and empirical validation of the Magnetic Sieve scaling law provide a foundational theoretical blueprint for utilizing magnetic topography as virtual nozzles in advanced plasma systems.

References

- [1] R. D. Richtmyer, *Taylor instability in shock acceleration of compressible fluids*, Communications on Pure and Applied Mathematics, vol. 13, no. 2, pp. 297-319, 1960.
- [2] T. Miyoshi and K. Kusano, *A multi-state HLL approximate Riemann solver for ideal magnetohydrodynamics*, Journal of Computational Physics, vol. 208, no. 1, pp. 315-344, 2005.
- [3] K. G. Powell et al., *A solution-adaptive upwind scheme for ideal magnetohydrodynamics*, Journal of Computational Physics, vol. 154, no. 2, pp. 284-309, 1999.
- [4] A. Dedner et al., *The essentially divergence-free system for ideal magnetohydrodynamics*, Journal of Computational Physics, vol. 175, no. 2, pp. 645-673, 2002.